

xv6 Assignment 3 — Changes Documentation

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1 List of Changes

Below are the changes made to the existing xv6 code base to implement the message passing system:

- **kernel/proc.h:** Added a `void* bufferpage` field to `struct proc` to store a reference to the process's allocated message buffer page.
- **kernel/proc.c:**
 - Implemented `msgenroll()`, which enrolls the calling process in the message passing system. It allocates a physical frame using `kalloc()`, zeroes it with `memset()`, maps it into the process's page table at the next available virtual address using `mappages()`, stores the virtual address in `p->bufferpage`, and updates `p->sz` to reflect the new allocation.
 - Implemented `msgsend(void* data, int size, int offset, int recipient)`, which sends data to a recipient process identified by `pid`. It loops through the `proc[]` array to find the recipient, converts the sender's data virtual address to a physical address using `walkaddr()` with a manual page offset correction, converts the recipient's buffer virtual address to physical, and copies the data using `memmove()`.
 - Implemented `msgread(void* data_out, int size, int offset)`, which reads from the calling process's message buffer at the given offset and writes the result to `data_out`. Both the buffer and output addresses are converted from virtual to physical using `walkaddr()` with manual page offset correction before copying with `memmove()`.